

CASE STUDY 5

Travel Agency with Two Branches

Designing a Wide Area Network for a two-office travel business

Group 5

CASE SCENARIO

A travel agency operates one main office and one branch office in another city.

Both offices exchange booking information, customer records, and financial reports daily — and need a network that keeps that exchange fast, reliable, and secure.

QUESTION 1 — NETWORK TYPE

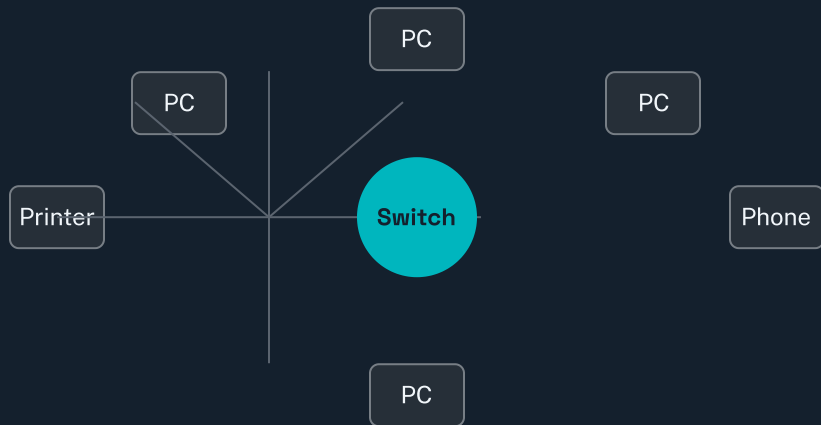
A Wide Area Network connects the two offices

A **WAN** is required to link two offices located in different cities. It overcomes distances that LAN or MAN technologies cannot reach, letting both locations operate as one connected network.



QUESTION 2 — TOPOLOGY

Each office uses a star topology

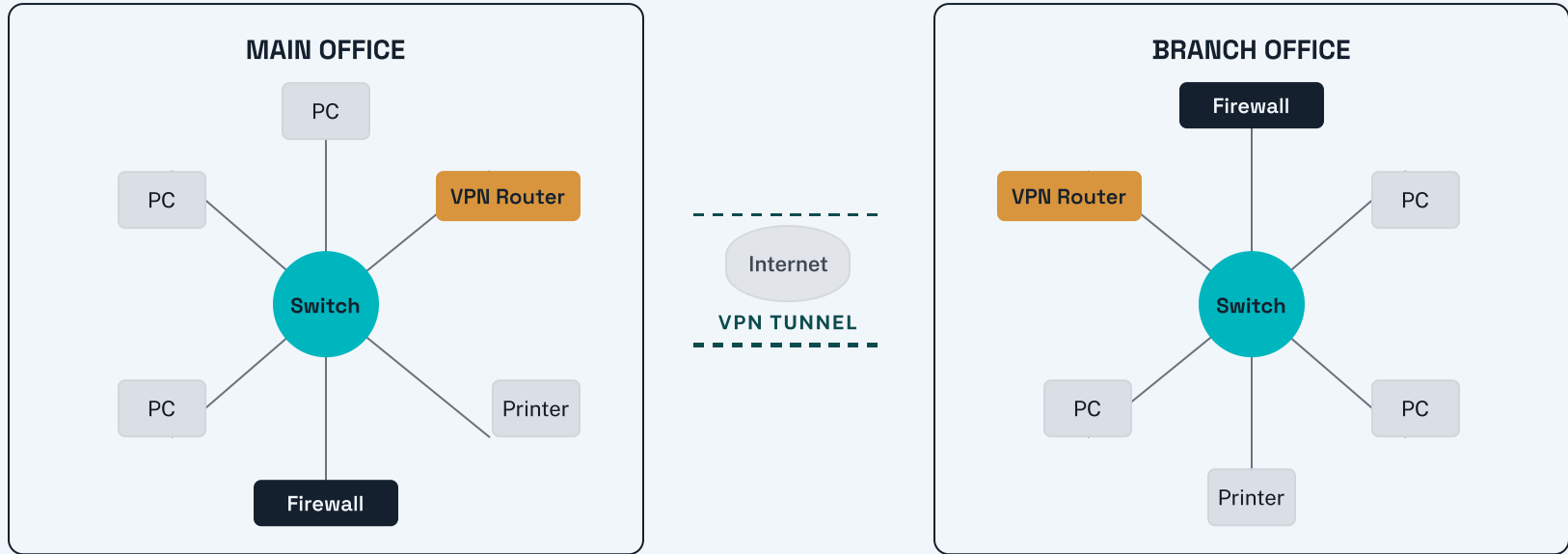


Every end device — computers, printers, phones — connects directly to a central switch.

- 01** High reliability — a broken cable or workstation only affects that one device.
- 02** Highly scalable — new agent workstations can be added as the business grows.

QUESTION 3 — NETWORK LAYOUT

Two star networks, joined by a Wide Area Network(WAN)



Main Office

Branch Office

QUESTION 4 — NETWORK DEVICES

What each office needs

Component	Qty	Purpose
VPN Routers	2	Facilitates inter-branch connectivity via a secure site-to-site VPN tunnel over the public internet.
Network Switches	2	High-density Ethernet ports managing local traffic within each office's star topology.
Next-Gen Firewalls	2	Deep packet inspection and security policies to safeguard customer and financial data.
End Devices	Variable	Agent workstations and IP-enabled peripherals for daily business operations.
Cat6a Cabling	Variable	Physical transmission medium for high-bandwidth LAN communication within each branch.

QUESTION 5 — BENEFITS

Why this design works

01

Enhanced Data Security

The site-to-site VPN keeps financial reports and customer records encrypted while moving between cities, preventing interception.

02

Fault Isolation & Availability

The star topology keeps localized hardware issues contained — a broken wire at one desk won't interrupt the rest of the branch.

03

Centralized Resource Sharing

Branch employees can securely access main-office databases and applications as if they were in the same building.

CONCLUSION

A WAN with an internal star topology gives the travel agency a secure, reliable, and scalable infrastructure.

The site-to-site VPN tunnel guarantees confidentiality for daily financial data and booking records, while the star topology minimizes downtime through excellent fault isolation — bridging the gap between the main office and its branch.